Pipeline Flow Chart

MEG/EEG Preprocessing Pipeline — Stages & Flow

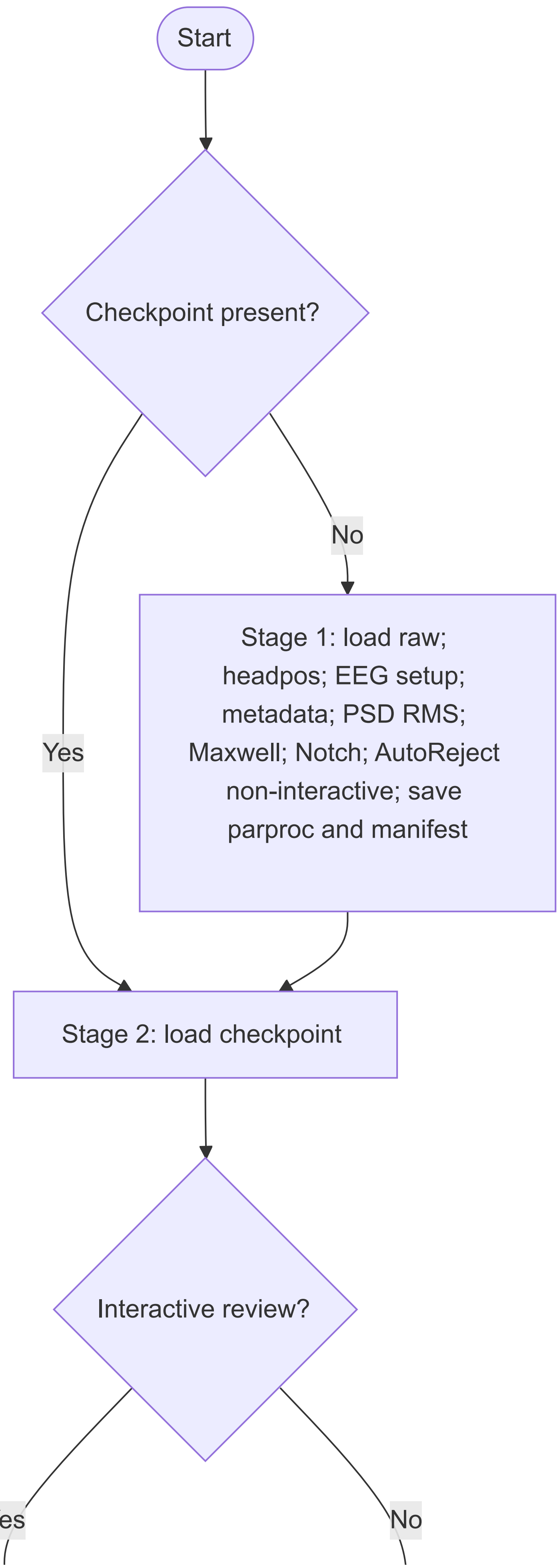
Overview

This pipeline runs in two main stages, with optional **hybrid HPC + local (MBPro)** mode to handle long-running computations and interactive review efficiently.

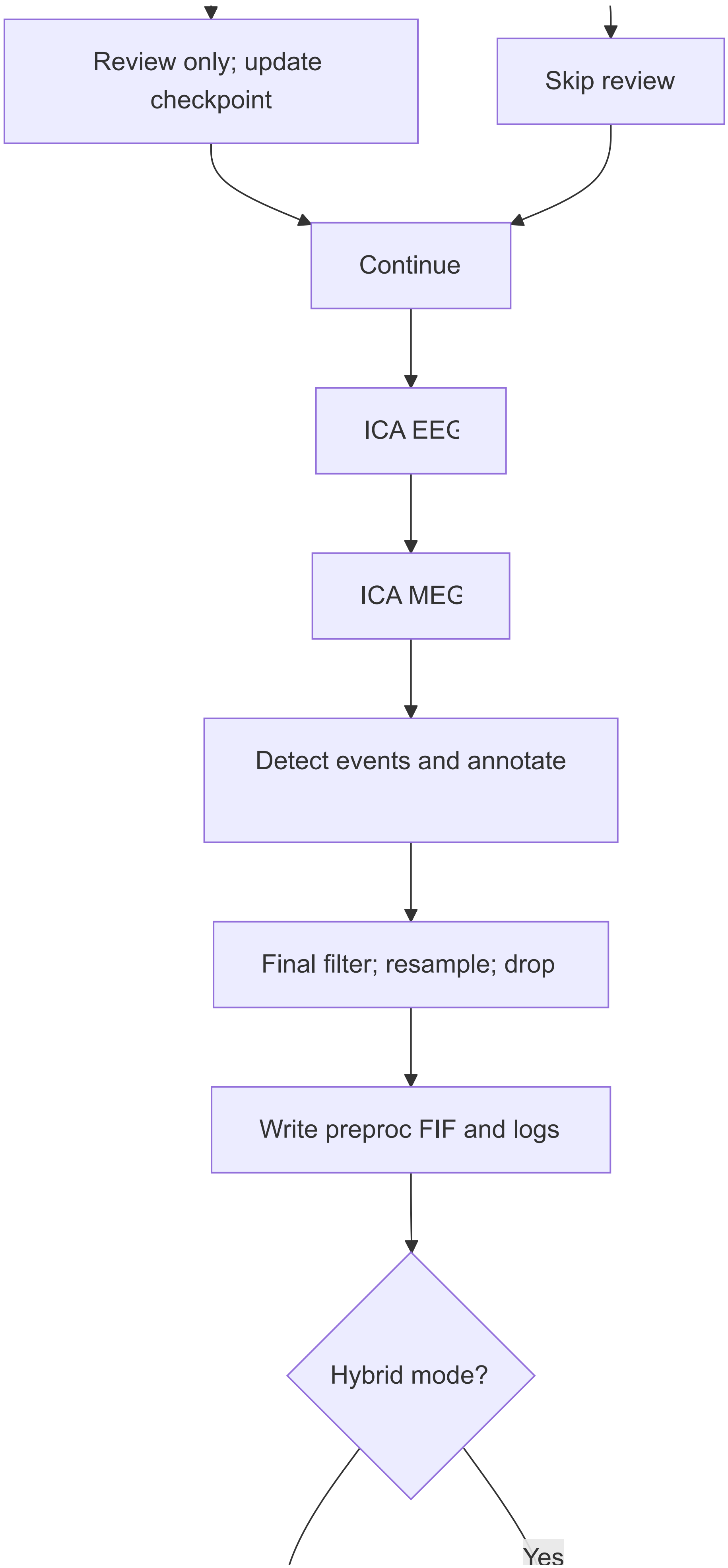
- **Stage 1** — Heavy computations; can run on HPC in headless mode.
- **Stage 2** — Interactive review and finalization; runs locally for responsive graphics.

Hybrid mode uses the **HybridTransferManager** to transfer raw and derivative files between HPC and local ./temp staging.

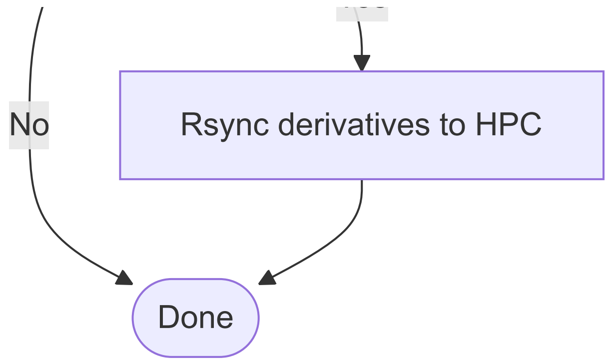
Flowchart



Start
Checkpoint present?
Yes
No
Stage 1: load raw; headpos; EEG setup; metadata; PSD RMS; Maxwell; Notch; AutoReject non-interactive; save parproc and manifest
Stage 2: load checkpoint
Interactive review?
Yes
No



Review only; update checkpoint
Skip review
Continue
ICA EEG
ICA MEG
Detect events and annotate
Final filter; resample; drop
Write preproc FIF and logs
Hybrid mode?
Yes



Stage 1 Modules

Step	Module / Function	Description
Load Config	run_pipeline() / detect_pipeline_stage()	Loads YAML, decides Stage 1 vs 2
HPC Prefetch	HybridTransferManager.fetch_all_bids_data()	In hybrid mode, rsync raw + derivatives to ./temp
Load Raw Data	read_raw_bids_robust()	Reads MEG/EEG data from BIDS
Head Position	get_head_pos_for_maxwell()	Compute or load head pos data
EEG Setup	prepare_eeg_channels()	Set montage, drop bad EEG
Metadata Repairs	apply_metadata_repairs()	Apply YAML-specified fixes
QC Diagnostics	qc_meg_raw(), plot_psd_and_peaks()	Pre-filter signal inspection
Maxwell Filter	apply_maxwell_filter()	tSSS and optional HPI correction
Notch Filter	raw.notch_filter()	Line noise removal
AutoReject (non-interactive)	run_autoreject_with_checkpoint()	Detect bad channels/epochs, save parproc checkpoint

Stage 2 Modules

Step	Module / Function	Description
Validate Checkpoint	validate_checkpoint_integrity()	Confirm parproc + manifest

Step	Module / Function	Description
YAML Consistency	run_autoreject_with_checkpoint()	Warn or enforce match
Load parproc	mne.io.read_raw_fif()	Load checkpoint data
Interactive Review	_run_interactive_review_only()	GUI bad channel review
Rank Estimate	mne.compute_rank()	Estimate MEG/EEG data rank
ICA (EEG)	run_ica()	Run + review ICA components
ICA (MEG)	run_ica()	Same for MEG data
Post-ICA QC	plot_psd_and_peaks()	Inspect cleaned signals
Event Detection	bitwise_events()	Stimulus/response annotation
Final Filter & Cleanup	apply_final_filter_and_cleanup()	Optional drop channels/resample
Save Outputs	write_bids_robust()	Write preproc derivatives
Write Logs	save_yaml(), JSON dump	Save processing logs
Push Back to HPC	HybridTransferManager.push_results()	Hybrid mode upload